AGILE LAB FAT

QUESTION:SMART ELECTRICITY BILLING SYSTEM

-NAME:VARUN PARASANNAA

-REG NO:23MIS0359

Smart Electricity Billing System
Problem Statement:

Ensure electricity usage is calculated and billed correctly.
Tasks:

1)Develop billing calculation logic

2)Write JUnit tests for tariff calculations

3)Implement CI/CD pipeline with automated testing

• Set up version control using GitHub

Configure CI pipeline (e.g., Jenkins/GitHub Actions) to trigger on code commits Automate build process using Maven Integrate automated unit tests in pipeline

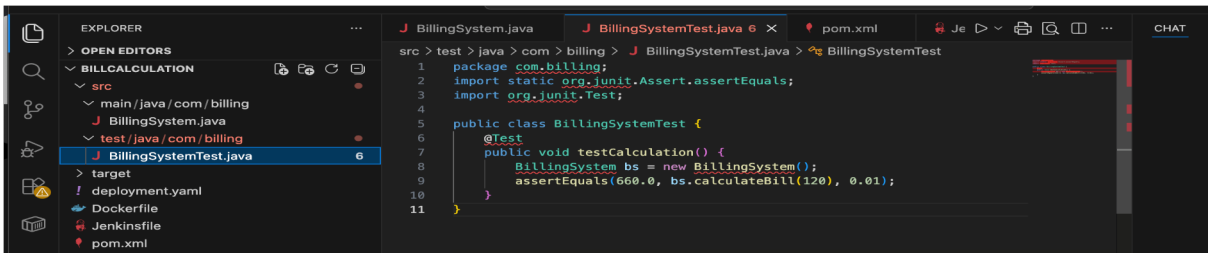
- Configure CD pipeline for automatic deployment to staging/production

4) Deploy services on Docker and Kubernetes

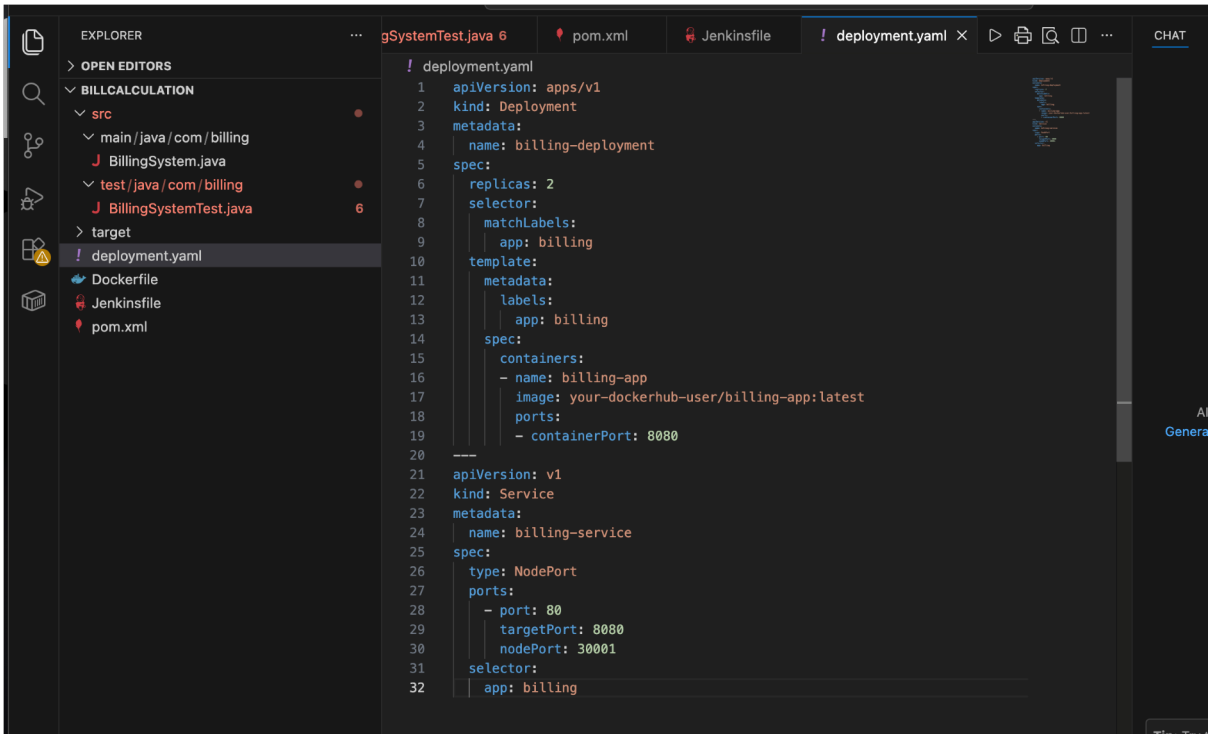
1) Develop billing calculation logic

Code:
Java code:

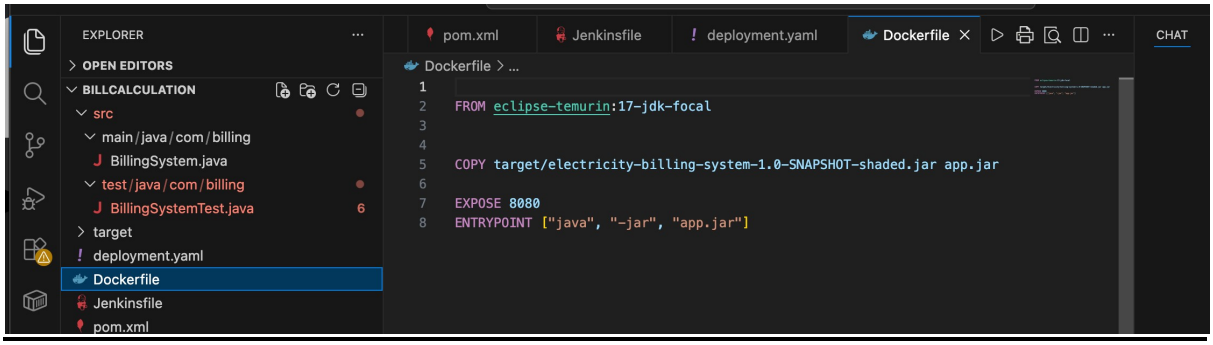
2)Junit test:



Deployment:



Docker:



3) Implement CI/CD pipeline with automated testing:

varunprasannaacvm / billcalculation

Type / to search

Code | Issues | Pull requests | Actions | Projects | Wiki | Security and quality | Insights | Settings

billcalculation Public

Pin | Watch 0 | Fork 0 | Star 0

main | 1 Branch | 0 Tags | Go to file | Add file | Code

varun prasannaa and varun prasannaa Initial commit: Billing system with Maven and Doc... 0a3ed98 · 5 minutes ago | 1 Commit

src	Initial commit: Billing system with Maven and Docker	5 minutes ago
target	Initial commit: Billing system with Maven and Docker	5 minutes ago
Dockerfile	Initial commit: Billing system with Maven and Docker	5 minutes ago
deployment.yaml	Initial commit: Billing system with Maven and Docker	5 minutes ago
pom.xml	Initial commit: Billing system with Maven and Docker	5 minutes ago

README

Add a README

Help people interested in this repository understand your project.

Add a README

About

No description, website, or topics provided.

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Contributors

No contributors

Languages

Java 88.1% Dockerfile 11.9%

Suggested workflows

Based on your tech stack

Jenkins / BillingSystem-Pipeline / #6

```
#8 unpacking to docker.io/library/billing-app:latest 0.0s done
#8 DONE 0.3s
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Kubernetes Deploy)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] sh
+ kubectl apply -f deployment.yaml
deployment.apps/billing-deployment created
service/billing-service created
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Deployment Successful!
[Pipeline] junit
Recording test results
[Checks API] No suitable checks publisher found.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Build Queue

No builds in the queue.

Build Executor Status 0/2

S	W	Name ↓	Last Success	Last Failure	Last Duration
✓		BillingSystem-Pipeline	14 min #6	16 min #5	58 sec

Pipeline script

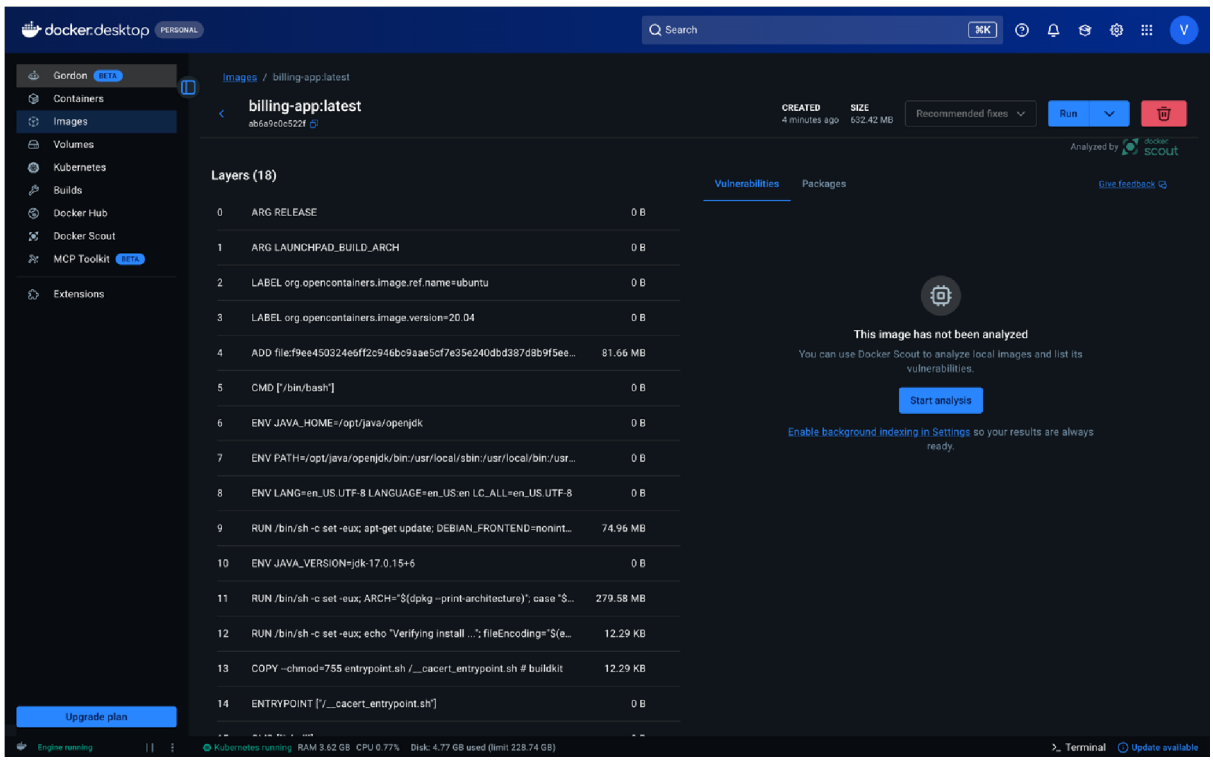
Script

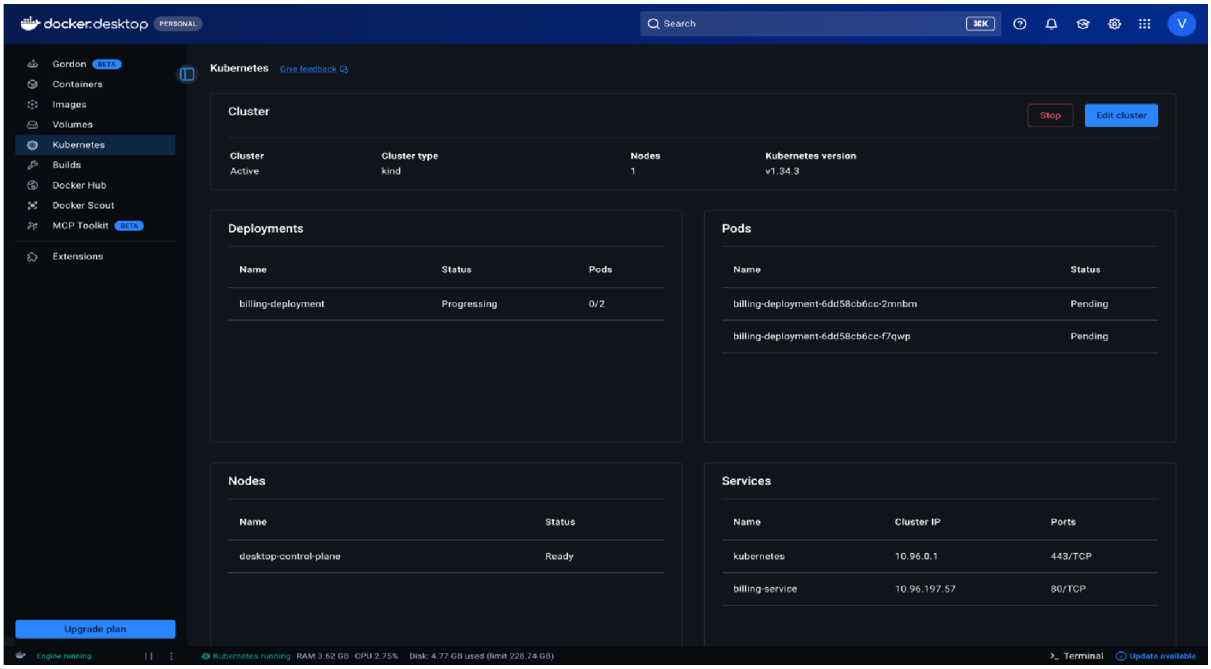
```
pipeline {
    agent any

    tools {
        // Must match the name in Manage Jenkins -> Tools -> Maven
        maven 'Maven3'
    }

    stages {
        stage('Checkout Code') {
            steps {
                // Pulls your code from GitHub
                git branch: 'main', url: 'https://github.com/varunprasannaacvm/billcalculation.git'
            }
        }
```

Use Groovy Sandbox

4. Deploy services on Docker and Kubernetes:

Images:






```
Last login: Wed Apr 15 16:31:22 on ttys000
varunprasannaa@varuns-MacBook-Pro ~ % kubectl get pods
No resources found in default namespace.
varunprasannaa@varuns-MacBook-Pro ~ % kubectl get service billing-service
Error from server (NotFound): services "billing-service" not found
varunprasannaa@varuns-MacBook-Pro ~ % kubectl get pods
NAME                                  READY   STATUS         RESTARTS   AGE
billing-deployment-6dd58cb6cc-2mnbm   0/1     ErrImagePull   0          2m2s
billing-deployment-6dd58cb6cc-f7qwp   0/1     ErrImagePull   0          2m2s
varunprasannaa@varuns-MacBook-Pro ~ % kubectl get service billing-service
NAME               TYPE       CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
billing-service    NodePort   10.96.197.57   <none>        80:30001/TCP   2m9s
varunprasannaa@varuns-MacBook-Pro ~ %
```